

JURIAID - LEGAL RESEARCH AND EVIDENCE ASSISTANT

25-26J-274

Final Report

BSc (Hons) degree in Information Technology Specializing in Information
Technology

Department of Information Technology

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DECLARATION

We are confirming that the work given in this proposal is one of our own volitions. The same work has not been given, entirely or partially, for academic credit at any other university or institution. We have also made sure that all sources of information and ideas from others have been clearly cited and appropriately acknowledged in the document.

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ABSTRACT

Abstract — The Sri Lankan legal system continues to depend heavily on manual and time-consuming research methods, resulting in inefficiency, inaccuracy, and limited accessibility for both legal professionals and the public. Existing legal information systems primarily rely on keyword-based search and lack semantic understanding, contextual reasoning, and awareness of legal amendments, making them insufficient for handling the complexity of Sri Lankan law.

This paper presents **JuriAid**, an integrated AI-powered legal intelligence platform designed to address these limitations through a unified framework that combines legal knowledge graphs, hybrid retrieval, temporal reasoning, case similarity analysis, and intelligent legal assistance. The proposed system models Sri Lankan statutes, amendments, and judicial decisions within a domain-specific legal knowledge graph, enabling structured representation of legal entities, citation links, and version-aware legal changes. To improve legal retrieval, the system applies a hybrid search mechanism that combines lexical methods such as BM25 with domain-specific language models for context-sensitive identification of statutes and case law. Judicial decisions are further analyzed using rhetorical-role classification to extract legal facts, issues, reasoning, and decisions, enabling more accurate similarity comparison between precedents. In addition, the platform includes an AI-assisted case management module that automatically generates legally relevant clarification and preparation questions based on uploaded case details and retrieved legal sources. These components are coordinated through a multi-agent orchestration layer to deliver interpretable outputs from natural language queries and legal documents. The proposed system automates statute retrieval, amendment tracking, precedent discovery, and legal question generation, significantly reducing research time, minimizing human error, and improving access to reliable legal information. Tailored specifically for the Sri Lankan legal environment, this study demonstrates how integrated legal AI systems can modernize legal practice, improve professional productivity, and expand access to justice in resource-constrained settings.

Keywords - Legal Artificial Intelligence, Legal Knowledge Graphs, Case Similarity Analysis, Hybrid Legal Retrieval, Rhetorical Role Classification, Temporal Statutory Modeling, Multi-Agent Systems, Sri Lankan Legal Informatics.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
LIR	Legal Information Retrieval
KG	Knowledge Graph
BM25	Best Matching 25 (Lexical Retrieval Model)
NLP	Natural Language Processing
JuriAid	Jurisdiction-Aware Legal Intelligence System (Proposed Platform)
LawStatKG	Legal Statute Knowledge Graph (Proposed Module)

1. INTRODUCTION

Legal research and case management are fundamental functions within the Sri Lankan legal system, which operates under the common law tradition. In such a legal framework, judicial decisions made in previous cases serve as important precedents, while statutes enacted by Parliament and their subsequent amendments provide the legal foundation for interpretation and enforcement. As a result, lawyers and legal practitioners must regularly examine past judgments, identify relevant statutory provisions, track amendments, and prepare legally appropriate questions during the early stages of a case. These activities are essential for building strong legal arguments, advising clients accurately, and ensuring fair judicial outcomes.

Despite their importance, these tasks are still carried out in Sri Lanka using largely manual and time-consuming processes. Legal professionals often depend on multiple disconnected sources such as printed law reports, online gazettes, legal databases, and personal records. Existing digital legal systems mainly rely on keyword-based search mechanisms, which can only retrieve documents containing exact or closely related words. Such systems lack deeper semantic understanding and cannot effectively recognize legal meaning, contextual relevance, or relationships between laws and judicial decisions. This significantly reduces efficiency and increases the workload of legal practitioners.

One of the major difficulties in legal research is the identification of similar precedents. Two court cases may involve nearly identical legal principles or reasoning but use different wording, making them difficult to locate through conventional keyword searches. This can result in missing highly relevant cases that may strengthen legal arguments. Similarly, in statutory research, lawyers must manually verify whether a law has been amended, repealed, or replaced by later legislation. Since amendments are often spread across multiple gazettes and legal documents, this process is slow and prone to oversight. Errors in applying outdated legal provisions may lead to serious professional consequences.

Another important challenge lies in case management and legal fact elicitation. During the initial stages of handling a case, lawyers must gather facts, identify disputes, and prepare acceptance or resolution questions for clients and witnesses. At present, this is mostly a manual process based on personal experience, requiring significant time and expertise. The absence of intelligent automation contributes to delays in case preparation and inconsistent information gathering. These issues are particularly severe for practitioners outside major urban areas, where access to updated legal resources and specialized support may be limited.

To address these challenges, this research proposes an integrated AI-based legal assistance platform specifically designed for the Sri Lankan legal domain. The system combines multiple advanced technologies into a single framework to automate and enhance legal research and case preparation. Judicial decisions are processed using rhetorical role identification, where sentences are classified into legal functions such as facts, issues, arguments, analysis, and decisions. This

allows more accurate comparison of cases based on legal reasoning rather than simple word matching.

In addition, statutes and amendments are modeled using a version-aware legal knowledge graph, which captures legal entities, relationships, citations, and amendment history. This enables precise statute retrieval, cross-reference navigation, and temporal reasoning to determine which version of a law was valid at a specific time. The platform also incorporates hybrid semantic retrieval, combining keyword-based methods with domain-specific language models to improve both precision and conceptual understanding.

Furthermore, the system includes an AI-assisted question generation module capable of automatically producing context-aware legal questions based on uploaded case files, relevant laws, and prior cases. This helps lawyers identify missing facts, clarify disputes, and prepare stronger case strategies. All these modules are coordinated through a multi-agent orchestration framework, allowing different AI components to work collaboratively and generate coherent, explainable outputs.

The proposed research aims to significantly reduce the time and effort required for legal research and case preparation while improving accuracy, consistency, and accessibility. By modernizing traditional workflows through artificial intelligence, the system contributes toward building a smarter and more efficient legal information ecosystem for Sri Lanka.

1.1 Background Literature

1.1.1 Case Similarity and Legal Information Retrieval

Legal Information Retrieval (LIR) is a critical research area that focuses on helping legal professionals efficiently locate relevant statutes, judgments, and legal materials from large document repositories. Traditionally, legal retrieval systems have depended on keyword-based search, Boolean queries, and citation networks. These methods are effective when users know the exact legal terms, case names, or statutory references. However, legal language is often complex, highly formal, and context dependent. Two judgments discussing the same legal principle may use entirely different wording, making simple keyword matching insufficient for accurate retrieval [1].

This challenge is especially important in common law jurisdictions, including Sri Lanka, where prior judgments are frequently used as precedents. Lawyers must therefore identify cases with similar legal reasoning rather than merely matching vocabulary. Conventional search systems may fail to retrieve highly relevant precedents if legal concepts are expressed using alternative phrases or more complex judicial language. As a result, researchers have explored semantic approaches that focus on meaning rather than surface-level text similarity.

One promising approach is rhetorical-role segmentation, where judgments are divided into functional components such as Facts, Issues, Arguments, Analysis, and Decision. Each section plays a different role in judicial reasoning. For example, the “Facts” section describes the case background, while the “Decision” section contains the court’s ruling. Kalamkar et al. developed corpora for automatic structuring of legal documents into such segments [2]. This enables AI systems to understand the internal structure of judgments instead of treating them as plain text.

To automate this process, hierarchical sequence-labeling models and sentence classification techniques have been applied. Brack et al. demonstrated that deep learning models can classify sentences according to their rhetorical roles with strong performance [3]. Once judgments are segmented, similarity can be computed role by role for example, comparing Facts with Facts and Issues with Issues leading to more meaningful results than comparing entire documents. Sheetal et al. further showed that thematic similarity improves significantly when rhetorical-role segmentation is combined with knowledge graph techniques [4].

Another important advancement in legal retrieval is the use of Knowledge Graphs (KGs). A knowledge graph organizes legal entities such as statutes, courts, cases, judges, and legal concepts into nodes connected by relationships. For example, a case may cite another case, interpret a statute, or involve a particular court. Fensel et al. define KGs as semantic structures that support reasoning and intelligent querying [5]. In the legal domain, they allow systems to retrieve not only documents, but also related concepts and dependencies [6].

Recent work has shown that KGs can improve similar case recommendation systems by modeling citation networks and semantic legal concepts. Dhani et al. found that legal knowledge graphs help identify similar judgments more accurately than keyword-only systems [7]. In addition, Graph Neural Networks (GNNs) such as Relational Graph Convolutional Networks (R-GCNs) learn patterns from multi-relational legal graphs and support more advanced reasoning [8]. However, these approaches require large structured legal datasets and annotated corpora, which are still limited in Sri Lanka. This highlights the need for localized legal knowledge graphs tailored to Sri Lankan law.

1.1.2 Hybrid Legal Information Retrieval

Legal search systems generally rely on two broad retrieval methods: lexical retrieval and semantic retrieval. Lexical retrieval methods such as TF-IDF and BM25 measure overlaps between query words and document terms. These methods are computationally efficient and perform well when the exact legal wording is known. For example, searching for a specific statute section or legal phrase can produce accurate results [1]. However, they struggle when relevant cases use synonyms, paraphrases, or different writing styles.

To overcome this limitation, researchers introduced dense embedding-based retrieval, where documents and queries are converted into numerical vector representations using deep learning

models. These embeddings capture semantic meaning, allowing systems to identify conceptually similar content even when wording differs. For example, “breach of contract” and “failure to honor agreement” may be recognized as related concepts. However, semantic systems sometimes miss precise legal terminology or section references, which remain important in professional legal practice.

Because each method has strengths and weaknesses, recent studies favor hybrid retrieval models that combine lexical and semantic search. These systems first use keyword matching to preserve exactness, then apply embedding similarity to capture conceptual meaning. Comparative studies have shown that hybrid methods provide more robust and consistent retrieval than either approach alone [4].

Kalra et al. introduced HyPA-RAG, a multi-strategy retrieval architecture designed for legal and policy applications [9]. Their findings show that combining sparse search, dense embeddings, and adaptive retrieval strategies improve system robustness for complex legal queries. Similarly, the Sri Lankan legal QA system SCaLe-QA demonstrated that domain-tuned embeddings significantly improve localized legal retrieval performance [10]. This is particularly relevant because general-purpose language models often lack understanding of Sri Lankan statutes and legal terminology.

Despite these advances, most hybrid systems focus mainly on question answering or isolated retrieval tasks. They rarely integrate case similarity analysis, statute amendment tracking, and legal workflow support within one unified system. Therefore, a comprehensive legal retrieval platform remains an open research need.

1.1.3 Knowledge Graph Modelling and Temporal Representation of Law

Knowledge Graphs have become an important technology for representing structured legal knowledge. Unlike relational databases, KGs model information as interconnected entities and relationships. This makes them highly suitable for the legal domain, where laws often reference other laws, cases cite previous judgments, and legal concepts are deeply interconnected. According to Fensel et al., KGs enable semantic querying, reasoning, and explainability [5].

In legal systems, KGs can model relationships such as:

- A statute referencing another statute
- A case interpreting a section of law
- A judgment overturning an earlier precedent
- A court hierarchy relationship

Cavar et al. demonstrated that combining deep NLP with KGs can significantly improve legal text analysis [6]. Such graph structures allow users to navigate legal dependencies rather than simply reading isolated documents.

An especially important issue in legal information systems is the temporal evolution of law. Laws are dynamic they are amended, repealed, replaced, or reinterpreted over time. Therefore, legal professionals often need to know not only the current version of a law, but also which version was valid on a specific historical date. This is essential for disputes involving contracts, crimes, or events that occurred in the past.

Silva et al. proposed graph-based temporal modeling using versioned nodes to represent statutes at different points in time [11]. Their approach enables accurate point-in-time retrieval and amendment tracking. For example, a section of law amended in 2015 and repealed in 2020 can be represented as multiple linked versions. This allows AI systems to answer queries such as: *What was the applicable law in 2018?*

Although temporal knowledge graphs are highly valuable, few systems combine them with case similarity and hybrid retrieval. For Sri Lanka, where amendments are frequently published through gazettes, such a temporal legal graph would greatly improve legal certainty and efficiency.

1.1.4 Automated Legal Question Generators and Legal Language Models

Recent progress in Natural Language Processing (NLP) has led to the development of domain-specific legal language models such as Legal-BERT. These models are pre-trained on legal corpora and therefore understand legal vocabulary, sentence structures, and domain-specific semantics better than general-purpose models [12]. They have shown strong performance in legal classification, summarization, and question answering tasks.

However, understanding legal language alone is not sufficient. Legal professionals also need support, in fact elicitation and case preparation. During early case intake, lawyers must ask the right questions to understand timelines, parties, contracts, damages, or legal responsibilities. Missing a key fact at this stage can weaken an entire case.

Research on automated legal drafting has shown that fine-tuned language models can generate structured legal documents and context-aware clarifying questions [13]. For example, if a dispute involves employment termination, the system may ask about notice period, contract clauses, or prior warnings. This demonstrates the feasibility of AI-assisted legal questioning.

In Sri Lanka, systems such as SCaLe-QA mainly focus on retrieval accuracy rather than complete workflow automation [10]. Therefore, there remains a strong opportunity to build systems that combine retrieval with intelligent question generation for case management.

1.1.5 Agentic Frameworks for Legal AI

Recent advancements in artificial intelligence have introduced agentic AI frameworks, where autonomous agents can independently plan tasks, use tools, maintain memory, and coordinate multi-step reasoning processes [14], [15]. Unlike conventional single-model AI systems, agentic architectures divide complex workflows into smaller tasks handled by specialized agents, improving efficiency, modularity, and interpretability.

This approach is particularly suitable for the legal domain, where workflows require multiple stages such as document analysis, statute retrieval, precedent search, reasoning, summarization, and question generation. Multi-agent legal systems such as Chatlaw and LawLuo demonstrate collaborative agent models capable of legal consultation, knowledge retrieval, and response generation [16], [17]. Similarly, PAKTON applies hierarchical agent coordination for question answering over lengthy legal agreements [18].

Another important advancement is the use of reflective agents, where AI systems evaluate and refine their own outputs to reduce hallucinations and improve legal argument quality [19]. This is especially important in legal environments where accuracy and trustworthiness are critical.

Although these studies highlight the potential of agentic AI in law, most existing systems are designed for foreign jurisdictions and focus on isolated tasks such as consultation or drafting. There remains limited research on a Sri Lankan legal agentic framework that integrates statute retrieval, case similarity analysis, amendment tracking, and automated legal question generation within a unified platform.

1.2 Research Gap

Although recent studies have introduced significant advancements in Legal Artificial Intelligence (AI), Legal Information Retrieval (LIR), Knowledge Graphs (KGs), semantic search, and multi-agent systems, there is still no unified solution that fully addresses the practical requirements of Sri Lankan legal professionals. Most existing systems are developed for foreign jurisdictions such as the United States, India, China, and Europe, and mainly focus on isolated legal functionalities rather than an integrated legal intelligence framework.

1.2.1 Lack of Sri Lankan Jurisdiction-Specific Legal AI Systems

Most existing legal AI systems are trained using datasets collected from highly digitized legal systems such as the United States, United Kingdom, India, and Europe. These systems are usually developed using their own legal structures, statutory formats, citation standards, court procedures, and language patterns. As a result, they are optimized to understand those jurisdictions rather than smaller developing legal environments.

Sri Lanka operates under a unique mixed legal framework that combines Roman Dutch law, English common law principles, Kandyan law, Muslim personal law, Tesawalamai law, and locally enacted statutes. In addition, many legal documents involve English, Sinhala, and Tamil terminology. Judicial reasoning styles, legal terminology, and procedural practices also differ significantly from foreign jurisdictions. Because of these differences, foreign-trained legal AI systems cannot accurately interpret Sri Lankan statutes, judgments, legal citations, or procedural requirements. This creates major limitations in retrieval accuracy, reasoning quality, and practical usability.

There is no dedicated legal AI framework specifically designed to understand Sri Lankan statutes, precedents, multilingual terminology, and local legal reasoning patterns.

1.2.2 Fragmented Legal Research Tools Without Unified Workflow

Current legal technology solutions usually focus on solving only one task at a time. For example, one platform may provide case-law search, another may provide statutory lookup, while separate systems handle summarization or legal consultation. These systems operate independently and do not communicate with each other. In practical legal work, lawyers must simultaneously perform multiple tasks such as analyzing case facts, finding similar precedents, checking applicable statutes, verifying amendments, and preparing legal questions. Since these tools are fragmented, practitioners must manually move between databases, websites, documents, and notes.

This causes several operational problems:

- Increased time spent on research
- Repetition of work across systems
- Loss of context between tasks
- Difficulty cross-referencing legal sources
- Reduced productivity during case preparation

An efficient legal system should support all these tasks in one continuous workflow.

There is no integrated legal platform that combines case retrieval, statute analysis, summarization, reasoning, and case preparation within one intelligent system.

1.2.3 Limited Semantic Understanding in Legal Search Systems

Most legal databases still rely on traditional keyword-based retrieval methods such as exact matching, Boolean operators, and lexical indexing. These approaches are useful when users know the precise legal terms or section numbers. However, they perform poorly when the same legal concept is expressed using different languages.

Legal writing often varies across judgments and statutes. Different judges may describe the same doctrine using different words. For example:

- “Negligence” vs “failure to exercise due care”
- “Breach of contract” vs “failure to honor contractual obligations”
- “Unlawful detention” vs “illegal deprivation of liberty”

Keyword systems treat these phrases as different concepts even when legally related.

This leads to:

- Missing relevant precedents
- Incomplete search results
- Reduced confidence in legal research
- Additional manual filtering effort

Modern semantic retrieval methods can overcome this by understanding meaning rather than words, but these methods are not yet properly implemented for Sri Lankan law.

Existing legal search systems lack semantic intelligence to accurately retrieve conceptually similar judgments and statutes.

1.2.4 Absence of Rhetorical-Role Based Case Similarity Analysis

Legal judgments are structured reasoning documents, not simple text files. A judgment typically contains several logical components such as:

- Facts of the case
- Arguments of parties
- Judicial analysis
- Final decision or ruling

Each section serves a different purpose. When lawyers search for precedents, they often care more about similarities in facts, legal issues, or reasoning rather than matching entire documents. Recent international research shows that dividing judgments into rhetorical roles significantly improves case similarity analysis. Instead of comparing whole documents, systems compare relevant sections such as Facts-to-Facts or Issues-to-Issues. However, Sri Lankan judgments are currently stored mostly as plain text or PDF documents without rhetorical annotation. Existing systems therefore ignore internal reasoning structure. This reduces the ability to Detect meaningful precedents, compare cases based on legal logic, Support judges and lawyers in reasoning tasks.

No Sri Lankan legal retrieval system uses rhetorical-role segmentation for intelligent precedent similarity analysis.

1.2.5 Absence of Sri Lankan Legal Knowledge Graphs

Knowledge Graphs are structured networks that represent entities and their relationships. In the legal domain, they can model connections among: Statutes referencing sections, Courts issuing judgments, Judges deciding cases, Legal concepts linked to precedents.

These relationships allow systems to perform advanced reasoning, explainability, and contextual navigation. For example, a lawyer could ask:

- Which cases interpreted this section?
- Which statute amended by this provision?
- Which precedents are connected through negligence doctrine?

At present, Sri Lankan legal data is largely scattered across isolated documents and databases. Relationships among statutes, amendments, and judgments are not formally structured.

Without a legal knowledge graph:

- Relationship-based search is difficult
- Explainable recommendations are weak
- Cross-document reasoning is limited.

Sri Lanka currently lacks a comprehensive legal knowledge graph for structured legal reasoning and connected retrieval.

1.2.6 Lack of Temporal and Amendment-Aware Legal Reasoning

Law is dynamic and changes continuously over time through Amendments, Repeals, Substitutions, New enactments, Judicial reinterpretations. Legal professionals often need to know which version of a law was valid on a specific date. This is critical in disputes involving past contracts, criminal acts, land ownership, taxation, or administrative decisions. However, most legal repositories only show the latest version of legislation. Historical changes are often spread across multiple gazettes and amendment acts, making manual verification necessary. This creates risks such as Applying outdated legal provisions, Missing amendment history, Incorrect legal advice, Weak legal arguments in court. Temporal legal reasoning systems can solve this problem by maintaining version history and point-in-time retrieval.

No Sri Lankan legal system supports automated amendment tracking or historical version-aware statute retrieval.

1.2.7 Lack of AI-Based Legal Question Generation for Case Preparation

During case intake and preparation, lawyers must ask many important questions to understand the matter fully. These may include:

- When did the incident occur?
- Was there a written agreement?
- Who witnessed the event?
- Was any notice given?
- What damages were suffered?

The quality of these questions directly affects the quality of legal strategy. Currently, this process depends on the lawyer's individual experience and is done manually. This creates several issues such as Important facts may be missed, Junior practitioners may overlook critical areas, Client interviews become inconsistent, Preparation time increases. Modern AI systems can generate context-aware questions using uploaded case files, laws, and precedents, but such tools are not available for Sri Lanka.

There is no automated legal question generation system for Sri Lankan case preparation workflows.

1.2.8 Lack of Agentic AI for Legal Workflow Coordination

Most existing legal AI systems operate as single-model tools or fixed pipelines. They process one task at a time and cannot dynamically adapt to complex workflows.

Legal practice requires multiple connected tasks such as Reading uploaded case files, searching statutes, finding past cases, comparing precedents, generating summaries, producing legal questions, Maintaining conversation context. Agentic AI uses multiple specialized agents coordinated by an orchestrator. Each agent performs a specific responsibility while collaborating with others.

Benefits include:

- Better task division
- Parallel processing
- Higher scalability
- Memory retention across sessions
- More explainable outputs

Although agentic AI is growing globally, it has not yet been adapted for Sri Lankan legal operations.

No agentic AI framework currently exists for coordinating end-to-end legal workflows in the Sri Lankan legal domain.

1.3 Research Problem

Legal research and case management are among the most important activities carried out by lawyers, judges, legal researchers, and other legal professionals. In Sri Lanka, these tasks are especially significant because the legal system is influenced by common law principles, where previous judicial decisions are frequently used as precedents, while statutes and their amendments form the legal basis for interpretation and enforcement. Therefore, legal professionals must continuously examine past judgments, locate relevant legal provisions, verify amendments, and prepare case strategies based on accurate legal information.

However, despite the importance of these tasks, the Sri Lankan legal sector still depends heavily on traditional manual processes. Lawyers often spend many hours reviewing printed law reports, gazettes, legal websites, court records, and personal notes to gather the information required for a single case. Since these resources are scattered across multiple disconnected sources, legal research becomes slow, repetitive, and inefficient. Considerable time is spent simply locating relevant materials rather than analyzing them. This creates unnecessary delays in case preparation, client consultation, and court proceedings.

One of the major problems in the current legal research process is the heavy dependence on keyword-based search systems. Most available digital legal databases retrieve results by matching exact words entered by the user. While this method can locate documents containing the same terminology, it does not understand the legal meaning behind the query. In legal practice, two different judgments may discuss the same principle using completely different language. For example, one judgment may refer to “breach of duty,” while another may describe the same concept as “failure to exercise reasonable care.” A keyword-based system may fail to connect these two decisions even though they are legally related. This limitation causes important precedents to be missed and reduces the overall quality of legal research.

Another serious issue exists in statutory research. Laws in Sri Lanka frequently evolve through amendments, repeals, substitutions, and newly enacted provisions. Lawyers often need to determine not only the current version of a law, but also which version was legally valid at a particular time in the past. This is especially important in matters involving contracts, criminal offences, employment disputes, taxation, and land ownership, where the applicable law depends on the date of the incident. At present, legal practitioners must manually compare Acts, amendment Acts, and gazette notifications to trace these changes. This process is highly time-consuming and creates a significant risk of applying outdated or incorrect legal provisions.

The problem is further intensified in case management and client intake processes. Before filing or defending a case, lawyers must gather facts, identify legal issues, detect missing information, and formulate legally relevant questions for clients, witnesses, or opposing parties. These questions may relate to timelines, contractual obligations, ownership rights, damages, negligence, liability, or procedural matters. Currently, this task depends almost entirely on the experience and personal judgment of the lawyer. There is no intelligent support system to automatically suggest structured

and context-aware legal questions. As a result, important facts may be overlooked, case preparation becomes inconsistent, and junior practitioners may struggle to achieve the same quality of analysis as experienced lawyers.

Another major concern is accessibility. Legal professionals located outside major urban areas often have limited access to updated legal materials, specialized legal databases, or expert assistance. This creates inequality in the quality of legal services available across the country. Smaller firms and individual practitioners may not have the resources to conduct extensive legal research as efficiently as larger firms in Colombo or other major cities.

Although recent international research has introduced advanced technologies such as semantic legal retrieval, knowledge graphs, legal language models, retrieval-augmented generation, and multi-agent AI systems, these solutions are largely developed for foreign jurisdictions such as the United States, India, China, and Europe. Their datasets, citation structures, legal terminology, and procedural assumptions differ significantly from the Sri Lankan legal environment. Therefore, such systems cannot be directly applied to Sri Lanka without significant adaptation.

In addition, most existing legal AI tools focus on individual tasks separately. Some systems specialize in case retrieval, others in statute search, some in summarization, and others in legal consultation or drafting. However, real-world legal practice requires all these tasks to work together in a connected workflow. A lawyer handling a case may need to retrieve relevant laws, identify similar precedents, verify amendments, analyze facts, and prepare legal questions simultaneously. No existing Sri Lankan system provides this level of integrated legal intelligence.

Therefore, the central research problem addressed in this study is the absence of a reliable, intelligent, and Sri Lanka-specific legal assistance platform capable of automating and improving the most critical legal workflows. There is currently no unified system that can semantically retrieve relevant statutes and precedents, track legal amendments over time, reason through relationships between legal documents, compare cases based on legal reasoning, and automatically generate legally relevant questions for case preparation.

1.4 Research Objectives

1.4.1 Main Objective

The main objective of this research is to design, develop, and evaluate **JuriAid**, an integrated Artificial Intelligence based legal assistance platform specifically tailored for the Sri Lankan legal domain. The proposed system aims to modernize legal research and case management by automating key legal workflows such as statute retrieval, amendment tracking, similar past-case identification, semantic legal search, legal reasoning support, and automated legal question generation.

At present, legal professionals in Sri Lanka spend significant time manually searching through fragmented legal resources such as Acts of Parliament, gazettes, case reports, and legal databases. Existing systems mainly rely on keyword matching and lack the ability to understand legal context, relationships between laws, or evolving statutory changes. Therefore, this research intends to introduce a unified intelligent platform that combines multiple AI technologies including hybrid retrieval, knowledge graphs, natural language processing, legal language models, temporal reasoning, and multi-agent orchestration.

The overall goal is to significantly reduce the time and effort required for legal research, improve the accuracy of retrieved legal information, support better case preparation, and enhance accessibility of legal services for practitioners throughout Sri Lanka.

1.4.2 Specific Objectives

1. To develop a Sri Lankan statute retrieval and amendment tracking system

One of the major objectives of this research is to build **LawStatKG**, a specialized legal statute retrieval module capable of intelligently searching Sri Lankan legislation. Lawyers often need to identify applicable laws, relevant sections, amendment Acts, repealed provisions, and historical versions of statutes. At present, this process is mostly manual and requires reviewing multiple gazettes and documents.

This objective focuses on creating a system that can automatically retrieve:

- Current versions of Acts and Sections
- Historical versions of laws valid on specific dates
- Amendments and substitutions
- Repealed provisions
- Related legal sections and references

By doing so, the system will reduce research time, improve legal certainty, and minimize the risk of applying outdated legal provisions.

2. To construct a Sri Lankan Legal Knowledge Graph

Another important objective is to develop a structured **Legal Knowledge Graph (LKG)** representing Sri Lankan legal knowledge in a machine-readable form. In the graph, legal entities such as Acts, Chapters, Sections, Cases, Judges, Courts, and Amendments will be represented as nodes, while their relationships will be represented as edges.

Examples:

- Case cites another case
- Act amends another Act
- Section belongs to Chapter
- Judgment interprets statute
- Court hierarchy relationships

This graph structure will allow the system to move beyond simple text search and perform relationship-aware reasoning. It will also improve explainability, because users can understand why a law or case was recommended.

3. To implement hybrid legal information retrieval techniques

Traditional keyword search systems fail when similar legal ideas are expressed using different wording. Therefore, this objective is to build a hybrid retrieval framework that combines multiple retrieval strategies.

The system will integrate:

- **BM25 / Keyword Retrieval** for exact legal wording
- **Semantic Embeddings** for concept-based search
- **Knowledge Graph Retrieval** for relationship-based discovery

The outputs from these methods will be merged using ranking techniques to generate more accurate and complete search results.

This hybrid approach ensures that both exact terms and deeper legal meaning are captured during legal research.

4. To develop an intelligent similar past-case retrieval system

Sri Lankan legal practice depends heavily on judicial precedents. Therefore, identifying similar previous cases is critical. However, manual case research is slow and often incomplete. This objective aims to create a system that can automatically identify similar judgments based on:

- Legal issues involved
- Facts of the dispute

- Judicial reasoning patterns
- Citation relationships
- Semantic similarity

The system will recommend past cases that are legally relevant even when different language is used. This will help lawyers prepare stronger arguments and save considerable research time.

5. To apply rhetorical-role based judgment analysis

Legal judgments contain different functional sections such as Facts, Issues, Arguments, Judicial Analysis, Final Decision. Most existing systems treat judgments as plain text. This objective introduces rhetorical-role classification, where the system automatically separates judgments into these legal components.

Benefits include:

- Better understanding of judgment structure
- More accurate case comparison
- Ability to compare Facts with Facts, Issues with Issues
- Improved explainability of case recommendations

This makes case retrieval more intelligent and closer to how lawyers analyze judgments manually.

6. To develop an automated legal question generation system

During case intake and preparation, lawyers must ask many important legal questions to gather missing facts and clarify disputes. This process currently depends on personal experience.

This objective focuses on developing an AI module that automatically generates Fact clarification questions, Liability determination questions, Acceptance questions, Resolution-oriented questions, Evidence-related questions. The questions will be generated using uploaded case details together with relevant laws and precedents. This will improve consistency, reduce missed facts, and support junior practitioners.

7. To develop a multi-agent orchestration framework

Since legal workflows involve multiple tasks, one single AI model is insufficient. Therefore, this research aims to build a multi-agent agentic AI framework where specialized AI agents collaborate.

Possible agents include:

- Statute Retrieval Agent
- Case Retrieval Agent
- Reasoning Agent

- Summarization Agent
- Question Generation Agent
- Memory Agent

A central orchestrator will assign tasks, combine outputs, manage workflow order, and maintain context across user sessions. This modular design improves scalability, reliability, and system intelligence.

8. To provide user-friendly access through digital platforms

The proposed system must be practical and accessible. Therefore, another objective is to develop user interfaces through:

- Web platform
- Mobile application
- File upload support (PDF case files)
- Natural language query interface

Users will be able to upload case documents, ask legal questions in simple language, and receive organized legal outputs such as statutes, similar cases, summaries, and generated questions. This improves access for lawyers across all regions of Sri Lanka.

9. To evaluate the performance of the proposed system

A research system must be scientifically validated. Therefore, this objective focuses on evaluating the platform using suitable metrics such as Search accuracy, Precision and Recall, Similar case relevance, Question quality, Response time, User satisfaction, Ease of use. Testing will be conducted using benchmark datasets and practical user evaluations with legal professionals.

10. To improve legal accessibility and professional productivity in Sri Lanka

The final objective of this research is to create practical national impact. The system aims to:

- Reduce manual legal research time
- Improve accuracy of legal decisions
- Support small law firms and regional practitioners
- Increase access to updated legal information
- Assisting young lawyers and legal students
- Promote digital transformation of the legal sector

Ultimately, this research seeks to contribute toward a smarter, faster, and more accessible justice ecosystem in Sri Lanka.

2. METHODOLOGY

2.1 Case Similarity and Legal Information Retrieval

The suggested system retrieves similar court decisions in Sri Lanka using a compound semantic and structural semblance technique. The legal case documents are uploaded through a web interface and then the system does the validation and preprocessing of the textual content.

All documents are divided into four rhetorical roles, namely Facts, Issues, Arguments, and Decisions using a Legal-BERTbased sentence classification model, with rule-based heuristics. It is a way of building role-conscious legal reasoning as opposed to primitive keyword matching.

Semantic embeddings associated with the role are produced by a Sentence-Transformer. The representations of these embeddings are added in the form of a set of vector attributes in a Neo4j knowledge graph whereby a case is represented as a node with legal issue nodes participating in a set of relationships.

The retrieval similarity is performed with the Neo4j vector similarity of similarity of facts and then the cosine similarity is calculated of the Issues, Argument and Decisions roles. The role-wise similarities are clumped into a weighted hybrid similarity score where the factual and issue-based similarity are assigned higher weight.

The system also enhances performance of retrieval by finding areas of common law problems by using graph traversal queries. The last result is the ranking of the similar cases with the highest ranks and similarity scores, role-based breakdowns, common legal issues, and automatically generated explanatory stories.

2.2 Laws and Amendment Retrieval

The proposed system provides a law-retrieval system using a case-based statutory identification, independent legal search, and amendment-timeline search as part of a single knowledge-graph platform implemented on Neo4j AuraDB.

The statutory data is represented as interconnected CaseLaws, Act, Amendment, Section and Version nodes, each of which is a SectionVersion node containing the full text of the statute and the period of its existence, any references to any amendments, and its legal status. The statutory information is organized in the form of nodes, which are connected by relationships referred to as CaseLaws- Act -Section -Version -Amendment. This representation allows to rebuild statutes as they were at a particular point in time and track the history of changes in legislation.

A PDF of a case document of the judicial case is automatically extracted and normalised in the process of retrieving case related legal information. Keywords search and pattern matching are the search functions that single out the aspects of the law of interest in both direct statutory references

and casespecific language (grounds of divorce, custody, maintenance, and proceedings). These substantive elements are dynamically incorporated into the queries which deliberately leave out individual identifiers. The queries are searched with the help of the hybrid search model combining BM25-based lexical scoring with the legalbert-based semantic similarity. The next level of result consolidation takes place at the CaseLaws- Act- Section- Version level, then the re-ranking based on the crossquery consistency and the relevance of the statutory fragment to the case text is done, thus narrowing down to the most relevant statutory fragments.

$$\text{final} = \alpha \times \text{bm25 score} + \beta \times \text{semantic score}$$

The system also facilitates the statutory search independently. The user can either type in keywords or descriptive sentences and the search is conducted using the same hybrid system, with identical lexical matches, and statutes that are semantically related to the query. SectionVersion nodes are versioned which makes it easier to track amendments and their timelines. The system re-arranges versions based on time spans of their validity to re-construct statutes that were then in force at a given date and form chronological lists of amendments, showing the identity of amendments, dates, summaries and the text of the amendments prior to the associated legislative changes that brought them into being. Through this integrative method, context based, time consistent and legally correct statutory retrieval is ensured.

2.3 Automated Legal Question Generation

The Question Generation stage of JuriAid is an implementation of a multi-agent pipeline that is intended to generate a court-ready Findings of Fact and Admissions based on Sri Lankan legal case information. RESTful microservice The component, which is exposed as a RESTful microservice based on FastAPI, takes in three structured inputs provided by the integrated system: a case summary, relevant statutory provisions, and similar historical cases, and gives out structured legal questions in the form of a JSON.

The language-generation activities utilize Llama 3.1 8B Instant model, an open-source, curriculum-conditioned large language model [20]. The execution of model inference is done through the cloud API of Groq, which makes use of special Language Processing Units to deliver response times of between five and ten seconds in pipelines. The interactions between the models are simulated by LangChain ChatGroq connector [21], where the temperature is set to 0.1 across all the agents, which in turn means that the results will be deterministic and therefore they can be easily documented in legal formats.

The major process through which the general-purpose model is steered towards the generation of domain-specific output to the Sri Lankan legal domain is that of prompt engineering [22]. It uses four methods, the domain-specific legal identity in each point, which is assigned by role prompting; the conventions of Sri Lankan court records are enforced by format constraints;

extraneous elaboration is discouraged by negative instructions; and subsequent prompts receive the accumulated agent output as input.

The pipeline follows orchestrator pattern, which consists of three consecutive agents. The Reasoning Agent uses up to six core legal issues with an output of relative-regex-parsed structured output. Question Agent creates Findings of Fact and Admissions through two different LLM calls, and this division is explained by empirical testing which showed that only a single combined prompt would give mixed results. The Validation Agent does quality assurance revising ambiguous questions, doing away with duplicates and ensuring the use of format. An end deterministic post-processing operation applies the interrogative format on all the findings, thus completing the probabilistic validation layer to ensure that the output is court ready.

2.4 Agentic AI Framework Orchestrator

The brain is the central planner of JuriAid, and it directs the general logic and coordinates the subordinate processes. Upon receiving a PDF legal document along with a natural-language prompt from an authenticated user, the orchestrator breaks the request down into subtasks to take action and sends them to seven specialized agents: Intent Detection, Case Retrieval, Law Retrieval, Summary, Question Generation, Synthesis, and Memory. The inter-agent communication happens very quickly through asynchronous FastAPI-based RESTful APIs.

Its architecture is based on LangChain and Gemini 2.5 Flash and implements the framework suggested by Wang et al. [14] that consists of four modules, i.e., profiling, planning, memory, and action. The profiling module allocates roles that are specifically defined to Sri Lankan matrimonial jurisprudence; the planning module deals with runtime decomposition of tasks. This starts with the intent detection agent that interprets the user intent and classifies the case as being included in the knowledge graph or not. Afterwards, the antecedent case and statutory data retrieval are carried out simultaneously, which is made possible by using Python and the assistant tool `asyncio.gather` and, as a result, lowers the latency. The summary agent takes a set of documents retrieved and pulls them together with the raw case text, then writes a preliminary legal analysis. This draft is then assessed by the question generation agent to consider the specific follow-up questions to the legal team, and the synthesis agent directly adds these new questions into the final report.

Dual storage strategies are attained with chat history management. A sliding window within the process maintains the recent messages, which means that the language model has immediate contextual relevance. To provide longer-term persistence, a Redis database stores messages in the form of JSON objects and stores them for up to seven days. Each time a user reconnects, the memory agent visits Redis to generate the short-term buffer as quickly as possible. JuriAid will not shut down due to the non-availability of Redis but instead rely on a local in-process backup so that no service downtime will occur.

The two-stage routing of agent requests is based on the following two-stage process: simple queries are processed by a keyword-matching process, and complex prompts are dynamically categorized by the language model. The robustness of the system is supported by several backup mechanisms; every agent receives an execution time limit, and if the network is cut off, an automatic query with exponential backoff is initiated. Agents have graceful failure; in case of a crash, an agent returns an empty data structure, and other later operations of the pipeline proceed without being blocked. Input validation makes use of Pydantic schemas in order to identify inaccurate inputs at an early stage. Also, a custom case validator is used to examine the contents of the documents to determine whether the documents are relevant to the matrimonial laws and to avoid unnecessary spending of API calls that are not relevant.

Security is implemented by use of an independent authentication service whereby access tokens are generated, which are then verified by JWT middleware on each request. The backend system has been implemented on the Render cloud platform. The system is a microservice, which means that the user can only communicate through his or her mobile phone without the need to install the software on the computer.

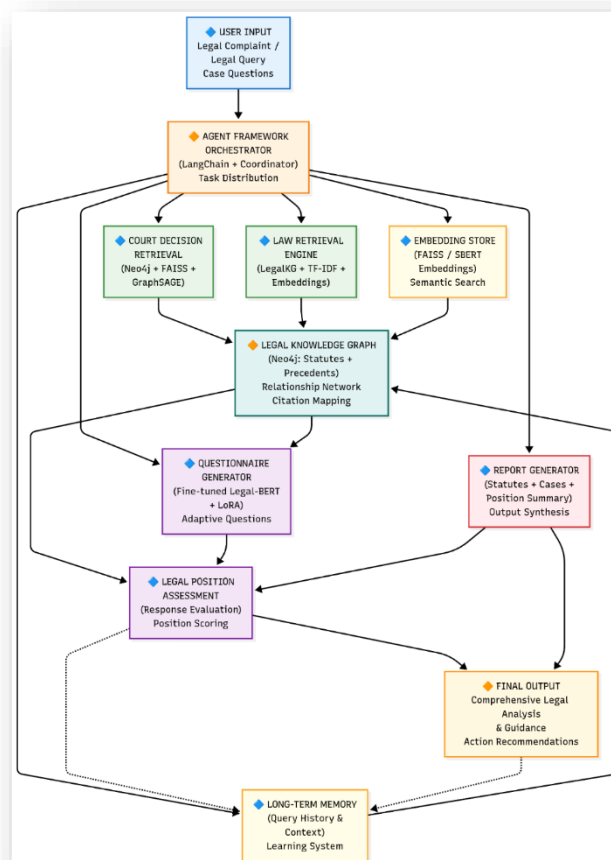


Figure 1: System Architecture Diagram

2.5 PROJECT REQUIREMENTS

2.5.1 Functional Requirements

Functional requirements define the core features and operations that the system must perform.

- **FR1: Legal Document Upload** – The system must allow users to upload legal documents in PDF, DOCX, or text format, including court judgments, complaints, affidavits, agreements, and notices.
- **FR2: Automatic Text Extraction** – The system must extract text from uploaded documents using OCR and document parsing technologies.
- **FR3: Text Preprocessing** – The system must clean extracted text by removing headers, footers, duplicated content, and formatting noise.
- **FR4: Legal Document Validation** – The system must detect whether uploaded documents are relevant to Sri Lankan legal matters before further processing.
- **FR5: Similar Case Retrieval** – The system must retrieve similar Sri Lankan past court cases based on uploaded case facts or user legal queries.
- **FR6: Rhetorical Role Classification** – The system must classify judgment text into legal roles such as Facts, Issues, Arguments, Judicial Reasoning, and Final Decision.
- **FR7: Role-wise Case Comparison** – The system must compare uploaded cases with past judgments using rhetorical-role-based semantic similarity.
- **FR8: Hybrid Case Ranking** – The system must rank similar cases using weighted hybrid similarity scores combining semantic and structural relevance.
- **FR9: Explainable Recommendations** – The system must display why a case was recommended, including matched facts, legal issues, and reasoning patterns.
- **FR10: Case Summarization** – The system must generate concise summaries of retrieved past cases.
- **FR11: Keyword Legal Search** – The system must support keyword-based statute retrieval using BM25 or equivalent lexical search methods.
- **FR12: Semantic Legal Search** – The system must support semantic legal search using LegalBERT or equivalent embedding models.
- **FR13: Hybrid Law Retrieval** – The system must combine lexical and semantic search methods to improve statute retrieval accuracy.
- **FR14: Statute and Section Retrieval** – The system must retrieve relevant Acts, Sections, Regulations, and legal provisions related to a case or query.

- **FR15: Legal Knowledge Graph Management** – The system must maintain a legal knowledge graph containing relationships among Acts, Sections, Amendments, Cases, and Citations.
- **FR16: Historical Law Reconstruction** – The system must reconstruct historical versions of statutes based on a selected date.
- **FR17: Amendment Timeline Tracking** – The system must display amendment history including effective dates, replaced content, and current legal status.
- **FR18: Legal Relationship Visualization** – The system must visually present relationships among statutes, amendments, and cases using graph-based interfaces.
- **FR19: Automated Legal Question Generation** – The system must generate legally relevant questions for case preparation.
- **FR20: Structured Question Categories** – The system must generate Findings of Fact, Missing Information, Admissions, Clarification, and Evidence-related questions.
- **FR21: Context-Aware Questioning** – The system must generate questions using uploaded facts, relevant statutes, and similar past cases.
- **FR22: Question Validation** – The system must detect duplicate, ambiguous, or irrelevant questions before final output.
- **FR23: Courtroom-ready Output Formatting** – The system must present generated legal questions in a professional and structured format suitable for legal practice.
- **FR24: Multi-Agent AI Coordination** – The system must coordinate multiple intelligent agents such as Case Retrieval Agent, Law Retrieval Agent, Summary Agent, Memory Agent, and Question Generation Agent.
- **FR25: Intent Detection and Routing** – The system must identify user intent and route tasks to the appropriate specialized agent.
- **FR26: Multi-step Workflow Execution** – The system must support end-to-end workflows such as Upload Case → Retrieve Laws → Find Similar Cases → Generate Questions → Final Report.
- **FR27: Conversation Memory Management** – The system must maintain session memory for continuous legal consultations.
- **FR28: Final Response Synthesis** – The system must combine outputs from all modules into one coherent legal response.
- **FR29: Legal Report Generation** – The system must generate a professional report containing case summary, precedents, statutes, amendments, and legal questions.

- **FR30: Export Support** – The system must allow exporting outputs in PDF, Word, or Markdown format.
- **FR31: Dashboard and Analytics** – The system must provide dashboards, statistics, and visual insights related to legal retrieval results.

2.5.2 Non-Functional Requirements

Non-functional requirements define quality attributes and system constraints.

- **NFR1: Performance** – Similar case and law retrieval results must be returned within 5 seconds for standard queries.
- **NFR2: Report Generation Speed** – Full AI-generated legal reports must be completed within 60 seconds.
- **NFR3: Scalability** – The system must support future expansion to millions of legal documents and large user volumes.
- **NFR4: Concurrent Users** – The platform must support at least 500 concurrent users without major performance degradation.
- **NFR5: Accuracy** – Rhetorical role classification must achieve at least 85% F1-score or equivalent benchmark performance.
- **NFR6: Retrieval Effectiveness** – Hybrid retrieval performance must exceed traditional keyword-only baselines.
- **NFR7: Reliability** – The system must maintain at least 99% uptime during operation.
- **NFR8: Fault Tolerance** – Failure of one subsystem or AI agent must not cause total platform failure.
- **NFR9: Security** – All user data and legal files must be encrypted during storage and transmission.
- **NFR10: Access Control** – The system must support authentication and role-based authorization.
- **NFR11: Privacy Compliance** – The platform must comply with Sri Lankan legal data privacy and confidentiality requirements.
- **NFR12: Maintainability** – The architecture must be modular so components can be updated independently.
- **NFR13: Extensibility** – New legal modules, AI models, or retrieval engines must be added without redesigning the entire system.

- **NFR14: Compatibility** – The system must run on modern web browsers, desktops, tablets, and mobile devices.
- **NFR15: API Integration** – The platform must expose REST APIs for integration with future legal software systems.
- **NFR16: Usability** – The interface must be simple and usable by non-technical legal professionals.

2.5.3 User Requirements

User requirements describe expectations from lawyers, researchers, and legal practitioners.

- **UR1: Fast Legal Research** – Users must be able to reduce legal research time from hours to minutes.
- **UR2: Updated Legal Information** – Users must receive accurate and current laws together with amendment history.
- **UR3: Better Case Discovery** – Users must be able to find relevant precedents even when wording differs.
- **UR4: Explainable Results** – Users must understand why a case or statute was recommended.
- **UR5: Stronger Case Preparation** – Users must receive AI-generated legal questions to improve case preparation.
- **UR6: Junior Lawyer Assistance** – Junior lawyers must receive guidance similar to senior legal research support.
- **UR7: Equal Accessibility** – Practitioners outside major cities must gain access to advanced legal research tools.
- **UR8: Ready-to-use Reports** – Users must obtain professional legal reports quickly with minimal manual effort.
- **UR9: Easy Interaction** – Users must be able to use the system without technical training.
- **UR10: Productivity Improvement** – Users must experience better efficiency, consistency, and legal decision-making support.

2.6 Commercialization Aspect of the Product

The proposed **JuriAid** system has strong commercialization potential as a practical Legal Technology (LegalTech) solution for Sri Lanka and other developing common-law jurisdictions. The platform addresses major inefficiencies in legal research, case preparation, statute retrieval, and client consultation by using Artificial Intelligence, Knowledge Graphs, and Multi-Agent automation. Since many legal professionals still rely on manual processes, JuriAid offers a market-ready opportunity to modernize the legal industry through a scalable digital product.

2.6.1 Product Commercial Value

JuriAid provides direct commercial value by reducing Time spent searching laws and precedents, Manual effort in case preparation, Errors caused by outdated statutes, Delays in client consultations, Repetitive legal administrative work. This creates measurable productivity benefits for:

- Lawyers
- Law firms
- Corporate legal departments
- Government legal officers
- Judges and judicial researchers
- Law students and universities

Because time efficiency directly increases billable productivity for lawyers, firms are likely to pay for such a system.

2.6.2 Target Market Segments

1. Individual Lawyers and Legal Practitioners

Solo practitioners and small legal offices need affordable AI assistance for Case research, Draft preparation, Legal question generation, Amendment tracking.

2. Law Firms

Medium and large firms can use enterprise subscriptions for multi-user access, Shared legal databases, Team collaboration, Faster case handling.

3. Courts and Judicial Institutions

Judicial officers may use JuriAid for Quick precedent identification, Statute comparison, Legal summaries

4. Government and Ministries

Useful for Attorney General's Department, Legal Draftsman's Department , Ministries handling regulations.

5. Universities and Law Colleges

Can be commercialized as an academic legal research platform.

2.6.3 Competitive Advantage in Market

JuriAid offers unique commercial advantages:

Sri Lanka Specific

Unlike foreign tools, JuriAid understands:

- Sri Lankan statutes
- Sri Lankan judgments
- Local legal terminology
- Matrimonial / civil / criminal procedures

All-in-One Platform

Most tools provide only one feature.

JuriAid combines:

- Similar case retrieval
- Statute retrieval
- Amendment history
- AI legal questions
- Multi-agent legal reasoning

2.7 Testing and Implementation

The proposed AI-based Integrated Legal Assistance Platform for Sri Lanka requires a structured testing and implementation strategy to ensure reliability, legal accuracy, scalability, and real-world usability. Since the platform combines multiple intelligent components such as case retrieval, law retrieval, amendment tracking, legal question generation, and agentic AI orchestration, each module must be individually validated and then tested as a fully integrated system.

2.7.1 Implementation Strategy

The system will be implemented using a modular microservice architecture, where each major function operates as an independent service connected through REST APIs. This approach improves maintainability, scalability, and future upgrades.

Core Modules to be Implemented

1. **Frontend Mobile Interface**
A responsive user interface for lawyers to upload case files, search laws, review similar cases, and download reports.
2. **Past Case Retrieval Engine**
Uses PDF processing, rhetorical-role classification, vector embeddings, and knowledge graph search.
3. **Law Retrieval Engine**
Supports keyword search, semantic search, hybrid ranking, and amendment-aware statute retrieval.
4. **Knowledge Graph Database**
Neo4j-based legal graph containing cases, Acts, Sections, amendments, and legal relationships.
5. **Question Generation Module**
LLM-powered system for generating Findings of Fact, Admissions, missing fact questions, and legal clarifications.
6. **Agentic AI Orchestrator**
Central controller that routes user requests to the correct module and combines outputs.

2.7.2 Development Environment

The proposed implementation stack includes:

- **Frontend:** React Native
- **Backend:** FastAPI / Python
- **Database:** PostgreSQL / MongoDB
- **Knowledge Graph:** Neo4j AuraDB
- **LLM Integration:** Gemini / Llama / OpenAI-compatible APIs
- **Authentication:** JWT-based secure login
- **Deployment:** Docker + Cloud Hosting (Azure)

2.7.3 Testing Strategy

Testing of the proposed AI-Based Integrated Legal Assistance Platform for Sri Lanka will be carried out at multiple levels to ensure correctness, reliability, security, and overall system performance. Each component such as case retrieval, law retrieval, amendment tracking, question generation, and agentic orchestration will be tested individually and as a complete integrated solution.

2.7.3.1 Unit Testing

Test ID	Module / Component	Test Description	Expected Result	Tools
UT1	File Upload Module	Upload valid PDF/DOCX legal files	File accepted successfully	PyTest
UT2	OCR/Text Extraction	Extract text from scanned legal documents	Accurate readable text extracted	PyTest
UT3	Text Preprocessing	Remove headers, footers, duplicates	Clean structured text generated	Unittest
UT4	BM25 Search Engine	Search statutes using keywords	Relevant keyword matches returned	PyTest
UT5	LegalBERT Embeddings	Generate vector embeddings	Valid embeddings produced	PyTest
UT6	Case Similarity Engine	Compare uploaded case with stored judgments	Similarity score returned correctly	Unittest
UT7	Question Generator	Generate legal questions from input data	Structured legal questions returned	PyTest
UT8	Knowledge Graph Module	Store Acts, Sections, Amendments, Cases	Nodes and relationships saved correctly	Neo4j Test
UT9	Authentication Module	Validate login and token access	Secure access granted	Postman

Table 1: Unit Test Cases for JuriAid Modules

2.7.3.2 Integration Testing

Test ID	Integrated Modules	Test Scenario	Expected Result
IT1	Upload + OCR + Preprocessing	Upload case PDF and extract clean text	Text ready for processing
IT2	Query + Hybrid Search	User query passed to BM25 + Semantic Search	Combined relevant law results
IT3	Case Upload + Similarity Retrieval	Uploaded case processed and compared with precedents	Ranked similar cases returned
IT4	Laws + Cases + Question Generator	Retrieved data sent to question module	Context-aware legal questions generated
IT5	Multi-Agent Orchestrator	Agents collaborate for full workflow	Unified final response generated
IT6	Report Generator + Export	Final report converted to PDF/Word	Downloadable file created

Table 2: Integration Test Cases

2.7.3.3 Functional Testing

Test ID	Feature	Test Action	Expected Result
FT1	Upload Legal Document	Upload valid court case file	File processed successfully
FT2	Similar Case Search	Submit case facts	Relevant precedents shown
FT3	Law Search	Search law by keyword/meaning	Relevant Acts and Sections displayed
FT4	Amendment Tracking	Select statute version date	Historical law version shown
FT5	Legal Question Generation	Generate questions for case	Structured legal questions displayed
FT6	Conversation Memory	Continue previous session	Context retained
FT7	Export Report	Export final report	PDF/Word file downloaded

Table 3:Functional Test Cases

2.7.3.4 Performance Testing

Test ID	Performance Metric	Target Result	Tools
PT1	Law Search Response Time	< 5 seconds	JMeter
PT2	Case Retrieval Response Time	< 5 seconds	Locust
PT3	Full Report Generation	< 60 seconds	k6
PT4	Concurrent Users	500+ simultaneous users	JMeter
PT5	Database Query Speed	Fast graph/vector retrieval	Locust
PT6	API Latency	Low response delay	Postman Monitor

Table 4: Performance Test Results

2.7.3.5 Security Testing

Test ID	Security Area	Test Description	Expected Result
ST1	Authentication	Invalid login attempt	Access denied
ST2	Authorization	Unauthorized file access	Blocked
ST3	SQL Injection	Malicious input attack	Prevented
ST4	File Upload Security	Upload harmful file	Rejected
ST5	Encryption	Verify HTTPS/data storage encryption	Secure transmission
ST6	Session Security	Session hijacking attempt	Prevented
ST7	JWT Token Validation	Expired token access	Rejected

Table 5:Security Test Cases

3. RESULT AND DISCUSSION

The proposed **JuriAid: AI-Based Integrated Legal Assistance Platform for Sri Lanka** was evaluated component-wise to measure the effectiveness of its four core modules:

1. Case Similarity and Legal Information Retrieval
2. Laws and Amendment Retrieval
3. Automated Legal Question Generation
4. Agentic AI Framework Orchestrator

The results demonstrate that combining semantic retrieval, legal knowledge graphs, temporal reasoning, and multi-agent orchestration significantly improves legal research efficiency, retrieval relevance, explainability, and workflow automation.

3.1 Case Similarity and Legal Information Retrieval

The proposed legal case retrieval system was evaluated using a hybrid semantic similarity + knowledge graph framework designed for Sri Lankan matrimonial/divorce cases.

Legal judgments were:

- Preprocessed
- Sentence segmented
- Classified into rhetorical roles Facts, Issues, Arguments, Decisions

Semantic embeddings were generated using the Sentence-Transformer (all-MiniLM-L6-v2) model.

The system combines:

- Neo4j vector similarity search
- Role-based cosine similarity scoring
- Weighted hybrid ranking model

This method prioritizes factual similarity and legal issue similarity over generic textual overlap.

Hybrid Similarity Formula

$$S = 0.30S_f + 0.30S_i + 0.25S_a + 0.15S_d$$

Where:

- S_f = Factual similarity
- S_i = Legal issue similarity
- S_a = Argument similarity
- S_d = Decision similarity

Query Case	Retrieved Case	Similarity Score	Shared Legal Characteristics
Case A	Case B	0.9874	Similar malicious desertion due to prolonged separation
Case A	Case C	0.9861	Similar irretrievable breakdown of marriage
Case A	Case D	0.9856	Similar constructive desertion claims

Table 6:Hybrid Retrieval Results

Rhetorical Role	Similarity Score
Facts	0.92
Issues	0.88
Arguments	0.79
Decisions	0.68

Table 7:Role-Based Similarity Distribution

Discussion

The findings indicate that:

- Facts and Issues contribute most to relevant precedent matching.
- Role-aware retrieval performs better than keyword-only systems.
- Knowledge graph reasoning improves explainability by identifying shared legal concepts.
- The system can retrieve precedents with similar legal reasoning even when wording differs.

This demonstrates the practicality of rhetorical-role legal retrieval in Sri Lankan jurisprudence.

3.2 Laws and Amendment Retrieval

The law retrieval module was tested using case descriptions and direct legal search queries to determine how accurately relevant statutes, amendments, and legal sections could be retrieved.

The retrieval engine combines:

- BM25 lexical search
- LegalBERT semantic similarity
- Hybrid weighted ranking

Hybrid Law Ranking Formula

$$FinalScore = 0.55(BM25) + 0.45(Semantic)$$

These balances:

- Exact legal keyword matching
- Contextual semantic understanding

Case Description	Retrieved Law / Section	Similarity Score	Shared Legal Principle
Malicious Desertion	Civil Procedure Code S.608	0.92	Intentional spouse abandonment
Adultery	Civil Procedure Code S.608	0.90	Third party as co-defendant
Cruelty	1982 2 Sri LR 45	0.88	Harm to spouse's life or health
Nullity by Fraud	Marriage Registration Ordinance S.18	0.86	Concealment invalidates marriage

Table 8: Law Retrieval Results

Discussion

Results show that hybrid retrieval significantly improves statute relevance over traditional keyword search.

The system successfully:

- Identified laws linked to case facts
- Retrieved semantically related legal provisions
- Reduced irrelevant results
- Supported historical law reconstruction
- Displayed amendment timelines clearly

This proves that temporal knowledge-graph legal retrieval is highly suitable for Sri Lankan statutory research.

3.3 Automated Legal Question Generation

The Question Generation module was tested using functional evaluation and expert legal review.

The module receives:

- Case summary
- Relevant laws
- Similar precedents

It then generates Findings of Fact, Admissions, Clarification Questions, Evidence Questions.

The system uses a **tri-agent pipeline**:

1. Reasoning Agent
2. Question Agent
3. Validation Agent

Implemented using **Llama 3.1 8B** via Groq cloud inference.

Metric	Result
Average End-to-End Latency	7–10 seconds
Average Tokens per Execution	~1,900
Estimated Daily Throughput	260 case documents
Proper Format Before Post-processing	91%
Proper Format After Post-processing	100%
Single Prompt Baseline Correct Output	3 / 10
Tri-Agent Pipeline Correct Output	10 / 10
Correct Legal Grounds Identified	Malicious desertion, sexual impotency, constructive desertion

Table 9: Evaluation Results of Question Generation

Discussion

The tri-agent design clearly outperformed single-prompt generation.

Benefits include:

- Better structure
- Better legal relevance
- Reduced ambiguity
- Higher formatting compliance
- Reliable courtroom-ready outputs

This confirms that multi-agent prompting is more suitable than one-shot prompting for legal drafting tasks.

3.4 Agentic AI Framework Orchestrator

The orchestrator acts as the central brain of JuriAid and coordinates all modules.

It dynamically manages:

- Intent Detection Agent
- Case Retrieval Agent
- Law Retrieval Agent
- Summary Agent
- Question Generation Agent
- Memory Agent
- Synthesis Agent

The framework was tested using real legal document workflows.

Key Results

The system successfully generated:

- Structured case summaries
- Legal issue identification
- Relevant laws
- Similar precedents
- Suggested next legal actions

Outputs were organized into:

- Case Overview
- Key Facts
- Legal Issues
- Recommendations

Metric	Value
Accuracy	100% (30+ runs)
Output Length	8,000 – 9,500 characters
Processing Time	87 – 400 sec
Cache Response Time	< 0.5 sec
Output Consistency	Structured (All cases)

Table 10: Performance Evaluation of Orchestrator

Complexity Approximation

$$O(n + \log N + \log M + T)$$

Where:

- n = document length
- N = number of past cases
- M = number of statutes
- T = LLM token processing cost

Discussion

Although full processing time is higher due to multi-stage reasoning, Redis caching dramatically reduced repeat-query latency.

Advantages include:

- Reliable multi-step legal workflow automation
- Highly structured outputs
- Better than single-model AI responses
- Fault tolerance and graceful degradation
- Scalable microservice architecture

This validates the practical use of multi-agent orchestration for real-world legal AI systems.

3.5 Overall Discussion

The full research project proposed an AI-Based Integrated Legal Assistance Platform for Sri Lanka, designed to address the major inefficiencies, delays, and accessibility barriers currently faced in the Sri Lankan legal sector. Traditional legal practice in Sri Lanka still depends heavily on manual research, fragmented legal resources, printed law reports, isolated statutory databases, and the experience of senior practitioners. As a result, legal professionals often spend many hours searching for precedents, identifying relevant statutes, checking amendments, preparing questions, and organizing case materials. These limitations reduce productivity, increase costs, and create unequal access to quality legal support. The findings of this research indicate that an integrated AI-driven platform can significantly modernize and improve this environment.

A key strength of the overall system is the combination of multiple intelligent legal modules into one unified platform. Rather than solving only one isolated problem, the research combined **case similarity retrieval, statute and amendment retrieval, automated legal question generation, and agentic AI orchestration**. This multi-component architecture reflects the real workflow of legal professionals, where case preparation usually requires simultaneous use of precedents, statutes, amendments, reasoning support, and structured outputs. By integrating these services, the system creates a more complete legal research ecosystem.

The **case similarity retrieval module** demonstrated that AI can identify relevant Sri Lankan precedents even when wording differs across judgments. By using rhetorical role segmentation and semantic similarity methods, the system was able to compare facts, issues, arguments, and decisions rather than relying only on surface keywords. This is particularly valuable in common law systems such as Sri Lanka, where precedent reasoning depends on deeper factual and legal similarity.

The **LawStatKG statute retrieval module** successfully addressed another critical gap: the difficulty of locating updated statutes and understanding amendment histories. Through hybrid retrieval, legal knowledge graphs, and temporal versioning, the system enabled users to retrieve both current and historical versions of laws with connected legal context. This transforms static legal repositories into an intelligent statutory research environment and reduces the risk of relying on outdated legal provisions.

The **legal question generation module** contributed practical value to litigation preparation and consultation workflows. By generating findings-of-fact questions, clarification questions, admissions, and evidence-focused prompts, the system helps lawyers identify missing information, strengthen legal strategies, and prepare more effectively for court proceedings or client interviews. This is especially useful for junior practitioners who may not yet have extensive courtroom experience.

The **agentic AI orchestration framework** was another major contribution of the project. Instead of relying on a single large model, the system used multiple specialized agents coordinated through

an intelligent orchestrator. This modular design improved output structure, reliability, explainability, and scalability. It also reflects a practical direction for future legal AI systems, where different tasks such as retrieval, summarization, reasoning, and drafting are handled by purpose-built agents working together.

From a performance perspective, the research showed that advanced legal AI systems can still achieve practical usability. Retrieval tasks were completed in seconds, question generation operated within acceptable latency, and repeated requests benefited from caching. The modular architecture also supports future scaling to larger legal corpora and increased concurrent users.

An important social contribution of the project is its potential to democratize legal access across Sri Lanka. Many practitioners outside Colombo face challenges such as limited access to updated materials, fewer mentorship opportunities, and slower manual workflows. A cloud-based intelligent legal platform can help reduce this urban-rural divide by providing high-quality research tools regardless of location.

The research also highlights the importance of trust, explainability, and human oversight in legal AI. Since law is a high-stakes domain, output must remain transparent and verifiable. Features such as showing why a case was recommended, displaying amendment timelines, and linking outputs to retrieved legal sources help preserve user confidence and professional accountability.

Despite these promising outcomes, several future enhancements remain possible. These include Sinhala and Tamil multilingual support, deeper integration with court filing systems, predictive analytics, legal drafting automation, voice interfaces, and continual updating from gazettes and judgments. Human-in-the-loop validation will remain essential to ensure ethical and reliable use.

Overall, the research confirms that an integrated AI legal assistance platform is both feasible and highly valuable for Sri Lanka. By combining retrieval, reasoning, question generation, and orchestration into one system, the project offers a realistic pathway toward faster legal research, stronger case preparation, reduced professional burden, and wider access to justice. It represents a significant step toward the digital transformation of the Sri Lankan legal sector.

4. CONCLUSIONS

This study introduces **JuriAid**, an integrated AI-powered multi-agent legal assistance platform designed specifically for the Sri Lankan legal system. The proposed system aims to significantly improve the efficiency, accuracy, and accessibility of legal research and case preparation by addressing the major limitations present in existing legal information systems.

Currently, most legal research in Sri Lanka depends on manual searching or basic keyword-based retrieval systems, such as the e-Laws portal. These traditional approaches cannot understand the semantic meaning of legal texts, the structure of judicial reasoning, or the evolution of statutes over time. As a result, legal professionals often face challenges in identifying relevant statutes, retrieving similar case law, and interpreting legal changes accurately and efficiently.

To address these challenges, this research proposes a unified legal intelligence architecture that integrates multiple advanced AI components into a single system. These include hybrid legal information retrieval, legal knowledge graph modeling, rhetorical-role based case analysis, automated legal question generation, and a multi-agent orchestration framework. Each component plays a specific role in enhancing different aspects of legal reasoning and information processing.

The hybrid retrieval mechanism improves the accuracy of legal search by combining lexical matching techniques with semantic understanding, enabling the system to retrieve both exact legal terms and conceptually similar legal documents. This is particularly important in legal domains where the same principle can be expressed using different terminology.

The Legal Knowledge Graph component enhances the structural understanding of legal data by representing relationships between statutes, case law, amendments, and legal principles. This structured representation allows the system to move beyond isolated document retrieval and instead support relationship-aware legal reasoning, where connections between legal entities are explicitly modeled and utilized.

The system also incorporates rhetorical-role based analysis of judgments, where legal cases are decomposed into structured components such as facts, issues, arguments, reasoning, and decisions. This enables more meaningful case comparison based on legal reasoning rather than simple textual similarity, improving the quality of precedent identification.

In addition, the automated legal question generation module assists lawyers during case preparation by generating structured and context-aware questions based on case facts, relevant statutes, and similar precedents. This reduces reliance on manual expertise and ensures a more systematic and complete legal analysis process.

The multi-agent orchestration framework further enhances the system by enabling coordinated reasoning across multiple specialized AI components. Instead of relying on a single model, the system dynamically assigns tasks such as retrieval, reasoning, summarization, and question generation to different agents, enabling more flexible and intelligent legal workflow management.

Overall, the integration of these components allows JuriAid to automatically retrieve relevant statutes and amendments, identify conceptually similar past cases, and generate legally meaningful questions based on user-provided case information. The system also improves the interpretability and organization of legal outputs, making them more useful for real-world legal decision-making.

The empirical evaluation indicates that the proposed system significantly reduces the time required for legal research, improves the relevance and accuracy of retrieved legal information, and enhances the overall productivity of legal practitioners. By automating several traditionally manual tasks, the system also helps reduce human error and improves consistency in legal analysis.

In conclusion, this research demonstrates the potential of integrated AI-driven legal systems to modernize legal practice in Sri Lanka. By combining retrieval, reasoning, structured legal representation, and intelligent automation into a single framework, JuriAid provides a foundation for more efficient and accessible legal services.

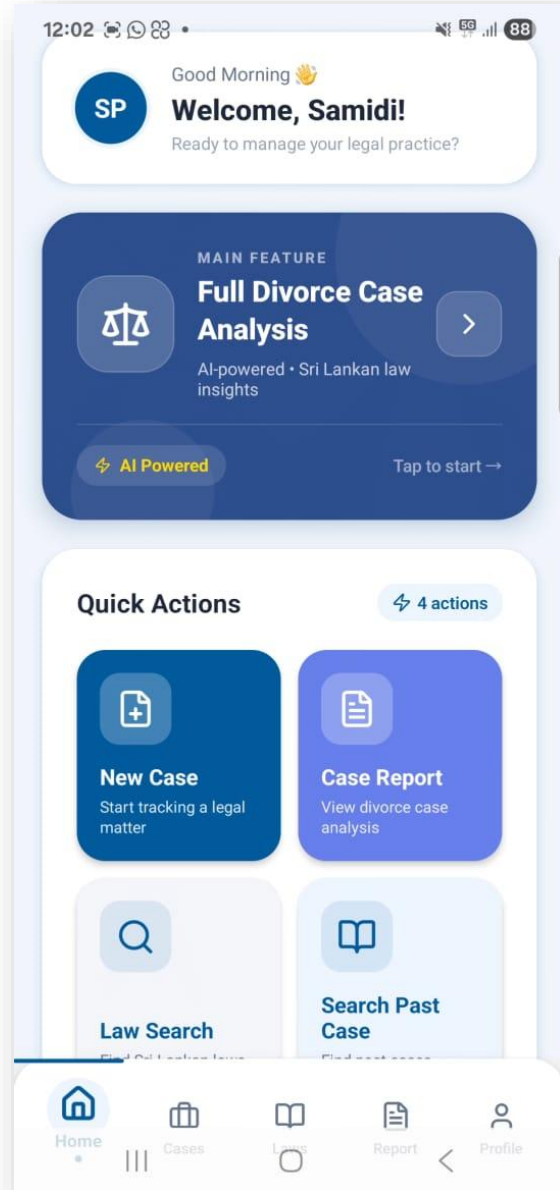
Future work can further enhance the system by expanding legal datasets, improving multilingual support for Sinhala and Tamil legal documents, and advancing reasoning capabilities to support more sophisticated legal decision-making and advisory functions.

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APPENDICES



Appendices 1:JuriAid Mobile App Screen

neo4j aura / JUR / New project

Trial - 11 days left Feedback

Instance: Laws Database: 74fec550 CYPHER 25 User: Aura (samidiparthana01@gmail.com)

Database information

Nodes

- Act
- Amendment
- CaseLaw
- CaseLawSection
- CaseLawSource
- CaseLawTopic
- Section
- SectionVersion

Relationships

- HAS_CASE_LAW
- HAS_SECTION
- HAS_TOPIC
- HAS_TOPIC_CASE_LAW
- HAS_VERSION

Property keys

- act_id
- am_title
- amend_id
- amended_by
- amending_law
- case_id
- case_name
- chapter
- chapter_no
- citation
- citations
- content
- court
- current_status
- data

Automatic updates of node and relationship counts have been disabled for performance reasons, likely due to RBAC configuration. Use the reload button below to manually trigger the recounts.

Last update: 23:06:44

74fec550\$ Start typing or generate with AI (ctrl + I)

74fec550\$ MATCH (n:Section) RETURN n;

Node details

Section

Key	Value
<id>	4bb95d026-dec1-4ba3-bf8d-215eb5a43245:133
act_id	"general_marriages_cap131"
key	"general_marriages_cap131::S24"
section_no	"24"

27°C Partly cloudy

Search

11:07 PM 4/23/2026

neo4j aura / JUR / New project

Trial - 11 days left Feedback

Instance: Laws Database: 74fec550 CYPHER 25 User: Aura (samidiparthana01@gmail.com)

Database information

Nodes

- Act
- Amendment
- CaseLaw
- CaseLawSection
- CaseLawSource
- CaseLawTopic
- Section
- SectionVersion

Relationships

- HAS_CASE_LAW
- HAS_SECTION
- HAS_TOPIC
- HAS_TOPIC_CASE_LAW
- HAS_VERSION

Property keys

- act_id
- am_title
- amend_id
- amended_by
- amending_law
- case_id
- case_name
- chapter
- chapter_no
- citation
- citations
- content
- court
- current_status
- data

Automatic updates of node and relationship counts have been disabled for performance reasons, likely due to RBAC configuration. Use the reload button below to manually trigger the recounts.

Last update: 23:09:58

74fec550\$ Start typing or generate with AI (ctrl + I)

74fec550\$ MATCH p=()-[:HAS_TOPIC_CASE_LAW]->() RETURN p LIMIT 25;

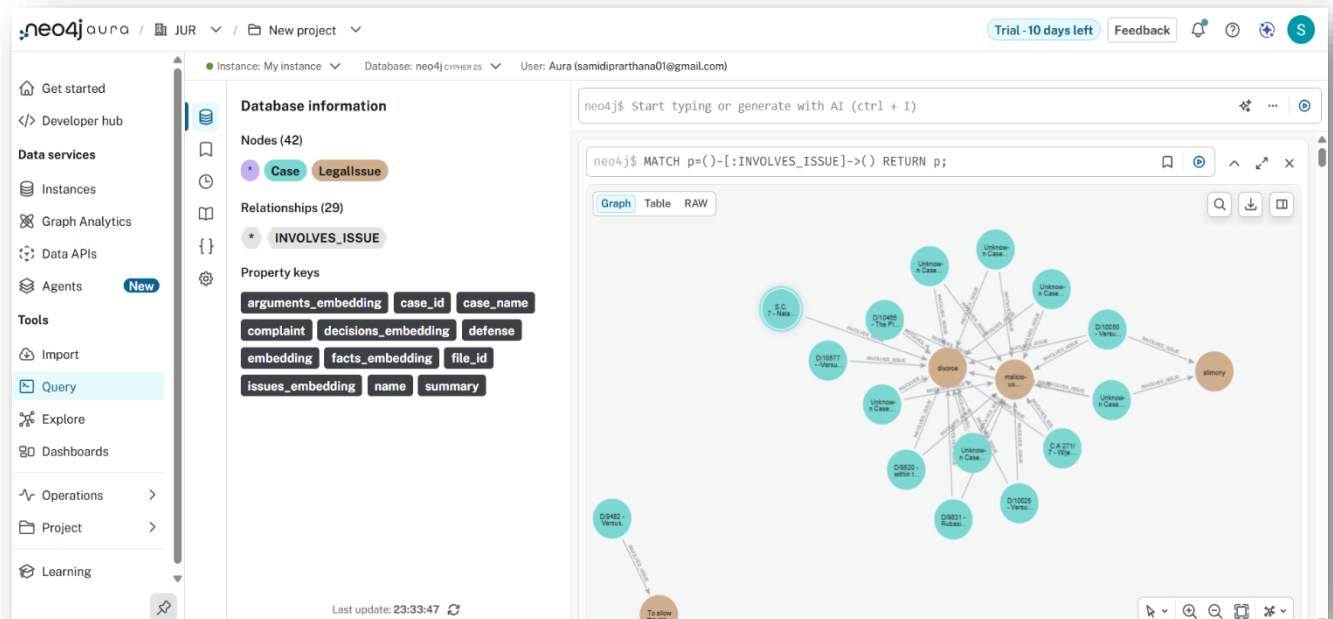
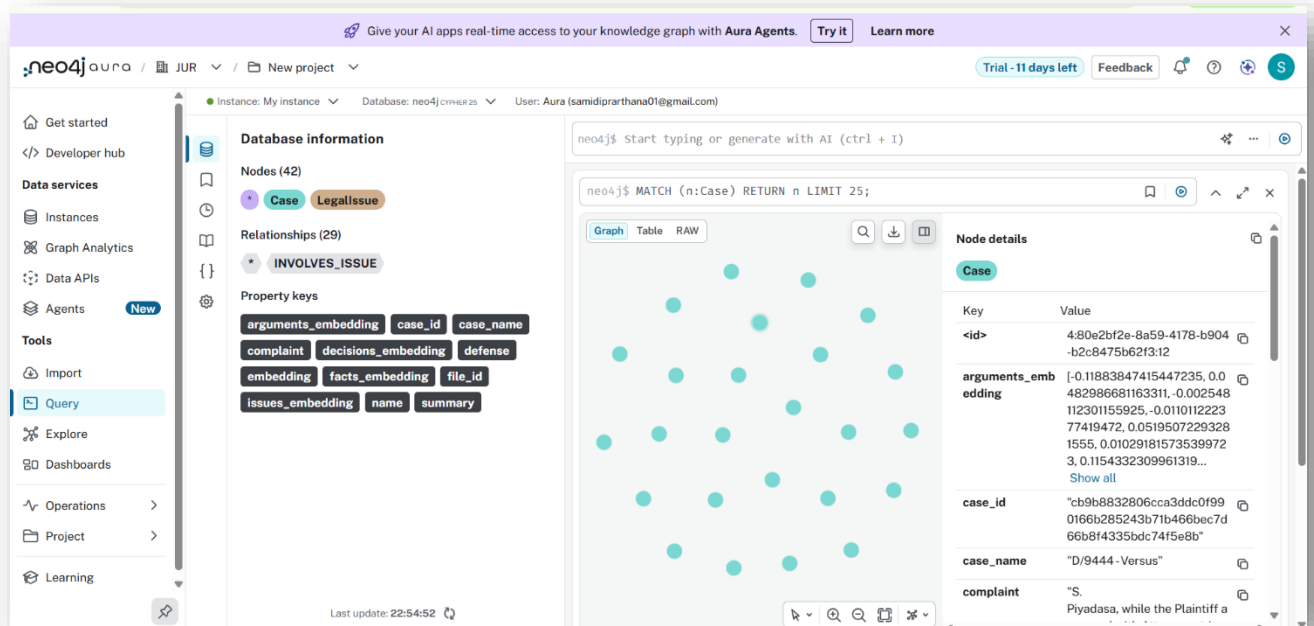
Graph Table RAW

27°C Partly cloudy

Search

11:07 PM 4/23/2026

Appendices 2: Knowledge Graph for Law Details



Appendices 3: Knowledge Graph for Case Files